# Post-Quantum Cryptography Migration

## Part 8 - Building & Shipping Software: Application Dev Stacks & Languages

**Compiled by the Spaced Research Ledger**

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The software that developers actually build with, web servers, language runtimes, database drivers, and browsers, mostly does not implement cryptography. It inherits it. When OpenSSL, Go, or the operating system's TLS stack turns post-quantum on, thousands of products above them change behavior at once.

This report follows that inheritance through four layers. Section 1 covers web servers and proxies, Section 2 the language runtimes, Section 3 database and messaging drivers, and Section 4 the browsers.

The inheritance is now mostly delivered for key exchange. Browsers have shipped hybrid post-quantum by default for over a year, and Edge has gone further, removing the off switch entirely. Servers built on OpenSSL 3.5, Go, or Apple's frameworks negotiate it without code changes. The visible gap runs along one seam: whatever sits on Windows or Java waits for opt-in settings or a September release.

The stragglers matter because they are load-bearing. Envoy, the proxy inside most service meshes, closed its post-quantum tracking issue as not planned. MongoDB documents nothing newer than OpenSSL 3.0. And signatures trail key exchange everywhere, with servers only now growing the code to select an ML-DSA certificate.

## 1. Web Servers & Proxies

*Web servers and reverse proxies terminate the world's TLS connections, so their support decides whether a website can actually speak post-quantum. Almost none of them implement TLS themselves. Each inherits from a cryptographic library, which makes this section a map of who links against what, and who has wired the new options through.*

### Recent Developments

The Windows side arrived last but arrived. IIS relies on the Schannel component, whose hybrid post-quantum support reached general availability in July 2026 [1]. Three hybrid groups are exposed, none enabled by default, and administrators must switch one on explicitly [1]. Only TLS 1.3 qualifies, with older versions permanently excluded [1].

Caddy, at the other pole, has key agreement on by default and waits only on Go's signature support [2]. Tomcat's pure-Java path is capped by the JDK, whose hybrid TLS lands with JDK 27 [3].

### Current State

The OpenSSL family inherits cleanly. Apache httpd gains support transparently through the library it links, with a one-line configuration directive to enable the hybrid group [4]. Researchers have built fully post-quantum Apache stacks for experimentation [5]. Open-source nginx picks the algorithms up from OpenSSL

3.5 [6], while the commercial NGINX Plus added support earlier through the add-on provider [7], covering whatever the provider offers [8]. Verification is a handshake check away [9], and Ubuntu's newest releases enable the hybrid group for nginx by default [10].

The Go and AWS-LC camps default on. Caddy shipped the standardized hybrid group by default in April 2025 [11]. It tracks the newest Go release specifically to inherit security features fast [11], and retired the pre-standard group as ML-KEM finalized [12]. HAProxy ships AWS-LC as its library, which carries the post-quantum curves natively [13]. Its maintainers closed a feature request as working as designed, treating post-quantum as the library's job [14].

Envoy is the notable holdout. Upstream Envoy does not support the standardized algorithms, and its tracking issue, which named the exact library commits needed, was closed as not planned [15]. Its TLS documentation does not mention post-quantum at all [16]. Tomcat, meanwhile, is building the signature side: recent releases select ML-DSA certificates from the client hello [17] and restrict TLS groups per host configuration [18].

They also layer post-quantum certificates onto classical contexts in the OpenSSL connector [19]. Windows completes the picture with certificate issuance now generally available beneath IIS [20].

### Unresolved Conflicts

Envoy's status depends on who you ask. Upstream closed its post-quantum issue as not planned [15]. Yet Red Hat's OpenShift Service Mesh, built on Envoy, announced configurable hybrid key exchange for its gateways and proxies in March 2026 [21].

A downstream build carrying newer BoringSSL would explain both, but neither source says so. Separately, a deployment guide asserts Envoy's library has shipped Kyber support since 2023 under the draft name, a claim in tension with the upstream issue [22]. A negotiated handshake against a stock Envoy build would settle it.

## 2. Language Runtime TLS Surfaces

*Most applications make secure connections through their language's standard library. Whatever Go, Java, .NET, Python, or Swift negotiate becomes what millions of programs negotiate. This section covers each runtime's TLS surface, where the pattern is a familiar one: key exchange is largely done, signatures are just beginning, and Java's calendar matters to everyone downstream.*

### Recent Developments

The runtimes converged within a few months of each other. Go 1.26 put all three hybrid groups in its defaults [23], and Go 1.27 added a standard ML-DSA package reaching into certificates and TLS [24]. Java's hybrid key exchange was delivered for JDK 27 [25], with backports planned to every long-term release through 2027 [26]. Windows Schannel, the stack beneath .NET's default TLS, went generally available in July, disabled by default and TLS 1.3 only [27][27].

Node.js now statically links OpenSSL 3.5, turning post-quantum on by default for its AWS connections [28], where the AWS SDK had supported hybrid key exchange since April [29]. Python's main cryptography package added native ML-KEM and ML-DSA, one pip install away [30]. Apple enabled hybrid key exchange by default across its 26-generation networking APIs [31], with the Swift server stack matching [32]. In C, wolfSSL went fully native [33] and its provider adds validated algorithms to any OpenSSL application [34].

### Current State

Go remains the cleanest story. Hybrid key exchange has been on by default since Go 1.24 whenever preferences are unset [35], a change that also removed the draft group [36]. The named constants are registered [37], with an escape hatch to revert [38]. The blemish is FIPS mode: the X25519 hybrid is listed as approved but always fails under the strictest setting [39], and the workaround disables the valid alternatives too [40].

Java is primitives now, TLS in September. The algorithms arrived in JDK 24 [41], with TLS integration deliberately deferred because the standards did not exist yet [42]. JEP 527 integrated into early-access builds [43], enabling the X25519 hybrid by default with no code changes [44], but general availability waits for the JDK 27 release [45].

.NET pairs native algorithms with an opt-in transport. .NET 10 ships all three standards in its cryptography namespace [46], as new types outside the classical base class [47], with Bouncy Castle remaining the cross-platform alternative [48]. C and C++ applications inherit from OpenSSL's defaults [49], on a long-term release supported to 2030 [49]. They can equally inherit from BoringSSL's earlier rollout in the Chromium world [50], or from liboqs through the provider [51]. Linux crypto policies steer the result on enterprise distributions [52].

The others range from transparent to vacant. Python's TLS behavior is entirely the linked OpenSSL's [53], with tooling to verify what was negotiated [53] and a hybrid-by-default research library at the fringe [54]. Rust's mainstream TLS ships the hybrid group behind a feature flag [55], superseding the experimental crate [56]. A survey finds core Rust crypto crates not prioritizing the standards [57], and wolfSSL's Rust support is still a roadmap item [58]. For Elixir, PHP, and Ruby, no established post-quantum information was found at all [49].

## 3. Database & Messaging Drivers

*Between applications and their data sit drivers and brokers, the connections that carry credentials and every query. Almost all of them borrow TLS from OpenSSL or a language runtime, so their post-quantum status is inherited, undocumented, or both. This section covers the data layer, where one database vendor has actually done the work.*

### Recent Developments

MySQL became the first major database to ship deliberate support. From version 26.7.0 it uses the post-quantum algorithms OpenSSL 3.5 provides, covering TLS 1.3 key exchange with optional signature advertisement [59], preferring hybrid groups by default with classical fallback [60]. The signature side is explicitly opt-in and does not conjure certificates [59]. One caution travels with it: key exchange is on by default, signatures are not, and collapsing the two into one support claim overstates the default [59].

Elsewhere the motion is inherited or requested. gRPC's C core defaulted to post-quantum key exchange in its July release [61], Kafka's ceiling tracks the JDK backport schedule [62], and Redis has an open feature request [63]. Postgres shows the edges: a client tool fails on ML-DSA server certificates that the command line accepts [64]. A JDBC security release fixed a channel-binding bug touching the new algorithms [65].

### Current State

Inheritance does the quiet work. Postgres's libpq has no TLS of its own, linking the system OpenSSL unchanged across versions [66]. Hybrid key exchange therefore works transparently in Python drivers once the library is new enough [67]. Redis has required OpenSSL for TLS since version 6 with the same dependency structure [68].

Kafka inherits from the JVM's TLS entirely [69], which means JDK 27's default hybrid groups reach it with no Kafka changes [70]. gRPC splits by language: the C core vendors BoringSSL [71], which shipped ML-KEM after standardization [72]. grpc-go rides Go's defaults, confirmed by its appearance in the Go FIPS bug report [73] [74].

The gaps are documentation-shaped. MongoDB's TLS pages reference nothing newer than OpenSSL 3.0 and no post-quantum group [75]. MySQL's own engineers had noted that upgrading the bundled library alone would not enable support, a plan the 26.7.0 release then fulfilled [76].

## 4. Browsers

*Browsers are where the migration touched the public first, and where it is furthest along. Every major engine now negotiates hybrid post-quantum key exchange by default, which is why most people are already protected without knowing it. This section covers the rollout, the disappearing off switches, and the one vendor whose story does not quite agree with itself.*

### Recent Developments

The rollout is entering its no-going-back phase. Chrome's default dates to version 124 with the draft group [77]. On Windows, the operating system beneath Edge gained its own configurable hybrid groups in July [78]. Apple's 26-generation release extends past the browser entirely, with the algorithms in its developer framework and even the iPhone-to-Watch link [79].

### Current State

Chrome set the pattern everyone follows. It shipped the standardized hybrid group as default in version 131 [80], and its enterprise override policy is deprecated with a version 146 end date [81]. Its certificate plan runs through Merkle Tree Certificates rather than traditional post-quantum X.509, with phases sketched into 2027 [82]. The measured result: over two-thirds of human TLS traffic to one large network now uses hybrid key exchange [83].

Edge mirrors Chrome and then leads it. The same default arrived at version 131 [80], the override policy died on schedule [84], and as of version 147 it is removed entirely, making post-quantum key agreement mandatory [85]. A separate WebRTC policy waits in the wings, not yet default [86].

Firefox covered the most surfaces. Desktop default came at version 132, behind a preference [80], extending to Android and to QUIC connections [80], and to WebRTC's handshake in version 146 [87]. Tor Browser inherits the default from its Firefox base [80].

Safari ships defaults with a compatibility warning. Connections on the 26-generation systems automatically advertise the hybrid group through the system networking frameworks beneath Safari [88]. Apple itself flags the risk: some legacy servers cannot handle the larger hello message and will fail to connect [88].

### Unresolved Conflicts

Apple's rollout description is self-inconsistent. Its support documentation says qualifying systems advertise the hybrid exchange automatically and uniformly [88]. An industry survey instead calls the Safari and WebKit rollout uneven across builds with no user-facing toggle [83]. Packet captures across current Apple builds would settle which is right.

## About This Report

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